

ReRule: Temporal Rule-Augmented Language Modeling for Causal Event Chain Completion

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Abstract

Complex web-based event sequences are essential for deciphering social dynamics and fostering positive social welfare outcomes computationally. Temporal knowledge graph completion and script event prediction have received attention for event sequence modeling. However, conventional knowledge graph models based on discrete quadruples fail to capture the global semantics and long-range dependencies of event chains, while script-based approaches often ignore the semantic roles and variations of entities within events. To address these limitations, we design a novel causal event chain completion task and propose a reasoning framework for key entity completion of query tuples in event chains. The framework first employs rule-guided filtering based on logical and temporal heuristics to prune the candidate space. Instruction-tuned generative models are then used to perform context-sensitive candidate generation and ranking. This hybrid design enables both factual consistency and flexible generalization. Our framework supports zero-shot reasoning as well as fine-tuned settings for domain-specific adaptation. We also construct three dedicated datasets for this task and conduct extensive evaluations. Experimental results demonstrate that our approach outperforms existing baselines across multiple metrics, highlighting its effectiveness in structured event modeling and entity inference.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Knowledge representation and reasoning**.

Keywords

Event Chain Completion, Rule Mining, Pre-trained Language Model

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1 Introduction

In the current digital landscape, web technologies have been deeply integrated into social structure and processes. Social media, news platforms, and various digital applications, while archiving global events in a timely and massive manner, are increasingly involved in shaping public awareness and social responses to major social issues, such as public health crises, climate change, and humanitarian relief. Therefore, the in-depth mining and analysis of these complex web-based event sequences constitute a critical pathway not only for deciphering social dynamics but also for leveraging computational methods to improve social welfare.

The Knowledge Graph Completion task (KGC) has attracted widespread attention in enhancing event knowledge modeling and reasoning. Traditional KGC methods primarily focus on static knowledge graphs (KGs), where facts are represented as triples (subject, predicate, object), assuming all information is temporally invariant. Numerous approaches have been proposed in this setting, including translational models such as TransE [1], which treat relations as vector translations, and more advanced methods such as RotatE [28], ComplEx [29], and ConvE [5], which leverage geometric or neural modeling to capture complex relation patterns. However, real-world knowledge is inherently dynamic and temporal in nature. To address this, temporal knowledge graphs (TKGs) extend static KGs by incorporating time annotations into each fact, resulting in quadruples (subject, predicate, object, timestamp). This formulation enables the modeling of entity evolution and relation drift over time. Temporal embedding-based methods such as HyTE [4], TTransE [11], and more recent models like ConT [19] and T-GAP [12] have demonstrated that time-aware projections or sequential encoders can significantly improve temporal link prediction. Moreover, attention-based and frequency-aware models such as ExKGR [35] and BoxTE [21] further exploit temporal locality and relational uncertainty to improve reasoning. Despite these advances, TKGs still struggle to capture complex narrative structures that span multiple related events. In many real-world scenarios, temporal events are not isolated, but unfold in semantically coherent sequences—referred to as *event chains*—that reflect underlying causality, coherence, and progression. Examples include

the escalation of international conflicts, multi-turn dialogues, and evolving user behaviors. Existing TKG models typically treat each event as an independent quadruple, thereby failing to capture the inter-event dependencies essential for narrative understanding. To mitigate this limitation, recent work has explored event-centric modeling techniques such as script learning and temporal event forecasting [16]. However, most of these methods lack structural grounding in TKGs or are limited in their generalizability to novel event contexts.

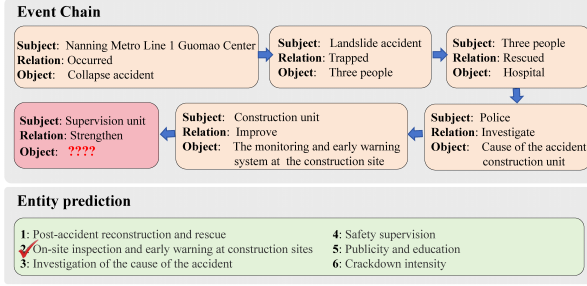


Figure 1: Causal Event Chain Completion.

To address these limitations, we propose a novel task *causal event chain completion*, where the objective of the task is to infer the missing entity—typically the object—in the final event of a temporally ordered event chain, as shown in Figure 1. Unlike conventional TKG completion tasks that consider isolated facts, our setting emphasizes context-aware inference over a sequence of interdependent events. It is highly relevant to practical applications such as event forecasting, narrative-based decision support, and early warning systems. This task poses unique challenges, including long-range temporal reasoning, representation of narrative flow, and handling of sparse or ambiguous targets. To address these challenges, we propose a hybrid inference framework that integrates symbolic reasoning with pretrained language models. Specifically, we incorporate (1) *rule-guided filtering* based on logical and temporal heuristics to prune the candidate space and (2) *instruction-tuned generative models* to perform context-sensitive candidate generation and ranking. This hybrid design enables both factual consistency and flexible generalization. Our framework supports zero-shot reasoning and fine-tuned settings for domain-specific adaptation. To validate the effectiveness of our method, we construct benchmark datasets from three sources: (i) ICEWS14, (ii) ICEWS05-15, and (iii) a self-curated Chinese temporal event dataset obtained via distant supervision and rule-based cleaning from real-world news corpora. These datasets across diverse domains and languages ensure a comprehensive evaluation of our model’s robustness and adaptability. The main contributions of this work can be summarized as follows:

- We formalize a novel task of causal event chain completion that bridges temporal KGs with narrative event modeling, enabling temporally grounded and context-aware entity inference.

- We propose a hybrid reasoning framework that combines symbolic rule-based constraints with instruction-tuned generative language models to improve accuracy and generalization.
- We build a multilingual benchmark using ICEWS14, ICEWS05-15, and a newly curated Chinese event dataset, and demonstrate that our approach achieves superior performance over strong zero-shot and fine-tuned baselines under standard evaluation metrics such as MRR and HIT@K.

2 ReRule

This section presents our proposed event chain entity completion model: **ReRule** (A multi-granularity rule-based and large language model-integrated event chain entity completion model). The overall architecture of the ReRule model is shown in Figure 2. The problem definition and each component of the event chain entity completion are then described in sequence.

2.1 Problem Definition

The entire event collection is represented as a unified temporal graph $\mathcal{G} = \{\mathcal{E}, \mathcal{R}, \mathcal{T}\}$, consisting of multiple temporally ordered event chains $\{C_1, C_2, \dots, C_m\}$. $\mathcal{E}$ is a entity set, $\mathcal{R}$ is a predicate set among entities and $\mathcal{T}$ is a time set. Each event chain C_j is a sequence of event triples $C_j = \{e_1, e_2, \dots, e_{n-1}, e_n\}$, where each event $e_i = (s_i, r_i, o_i)$ denotes a subject-relation-object triple. The final event $e_n = (s_n, r_n, ?)$ is the target query denoted as q , for which the object is missing.

Unlike prior approaches that directly learn an embedding for the missing object o_n , we propose an event chain entity completion method that integrates rule-based knowledge with large language models. Specifically, we construct a two-level rule set $\mathcal{R}_{\text{rule}}$ encompassing both concept-level and entity-level rules, and leverage LLMs to generate a candidate set of object entities $O_c = \{o_1^c, o_2^c, \dots, o_k^c\}$. The final objective is to identify the most likely entity $\hat{o}_n$ from the top-ranked candidates:

$$\hat{o}_n \in \text{TopK}(\text{LLM}(\mathcal{G}, \mathcal{R}_{\text{rule}})) \quad (1)$$

where $\text{LLM}(\cdot)$ denotes the generative module enhanced with rule-based prompts, and $\hat{o}_n$ is the predicted target entity.

2.2 Similar Event Component Search

Due to the limitations of NLP tools in removing redundancy in the novel constructed event chain completion datasets that are structurally complete and semantically coherent, semantically identical entities may appear with different surface forms (e.g., "a man" vs. "man", or "police", "officers", "110 personnel"), which introduces noise in entity prediction. To mitigate this, we employ the event representation calibration to obtain a consistent event representation for semantical events with diverse expressions. We first embed the entity set $\mathcal{E}$ and predicate set $\mathcal{R}$ using a BERT encoder, and use cosine similarity to compute pairwise similarity scores. Then, the scores between entities or predicates exceeding a similarity threshold are obtained globally as a candidate entity or predicate. Meanwhile, we apply semantic role labeling with LTP 2.0 [3] to construct coreference rules within each chain and merge co-referent mentions.

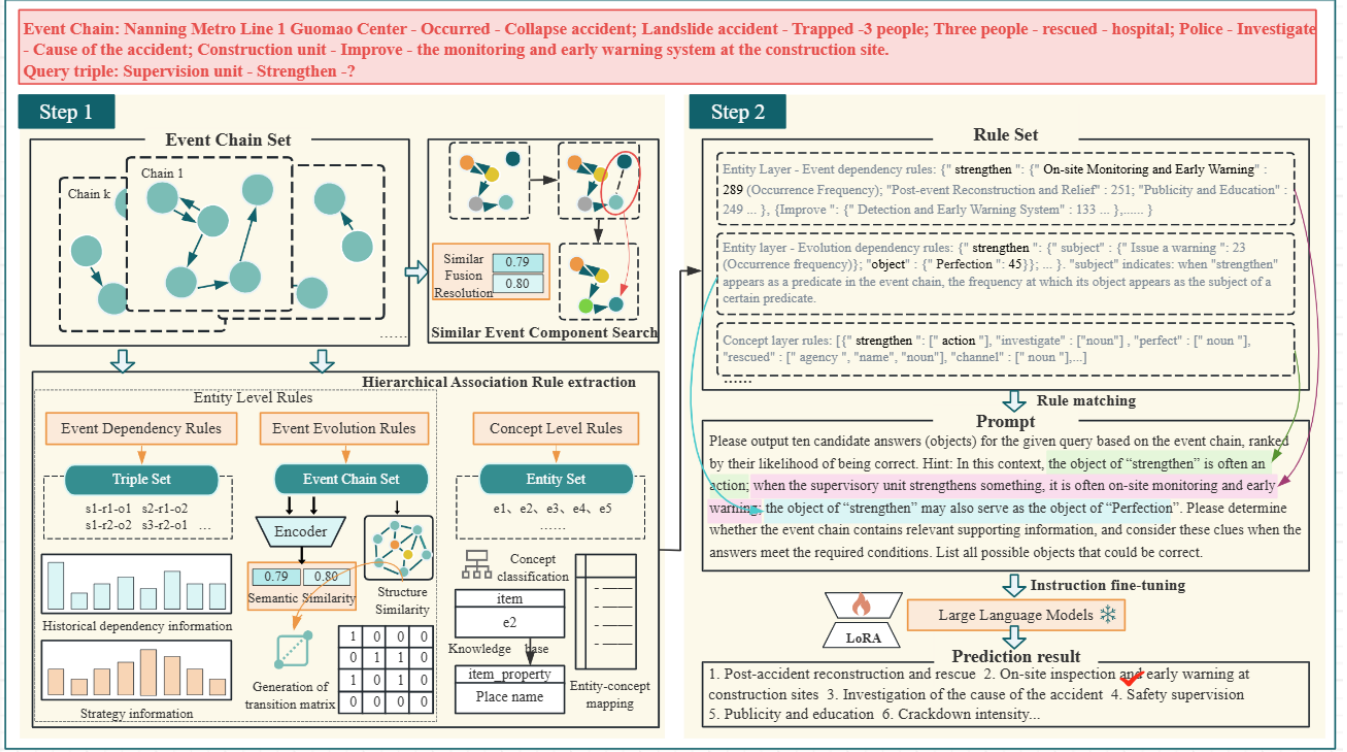


Figure 2: Architecture of the proposed ReRule model, integrating multi-granularity rules with large language models for event chain entity completion.

Furthermore, we reconstruct event chains into textual descriptions, encode them with a fine-tuned BERT model, and compute the similarity between chain representations. The resulting entity, predicate, and event similarity sets are then used in subsequent modules (e.g., dependency inference, rule generation) to expand sparse patterns and support robust reasoning semantically.

2.3 Event-level Dependency Inference

Many event triples, especially resolution-type predicates, commonly recur across chains. Thus, we employ the frequency of entities associated with the predicate to capture triple-level dependencies [34]. Although many events in knowledge graphs follow recurring patterns, some queries lack historical context. We therefore model both historical and non-historical frequencies. Specifically, for query $q = (s, r, ?)$, we compute the frequency of each entity co-occurring with (s, r) , and represent it as $Z^{(s,r)} \in \mathbb{R}^{|\mathcal{E}|}$:

$$Z^{(s,r)}(o) = \lambda \left(\Phi_{\sum |\{s|(s,r,o) \in \mathcal{L}'\}| > 0} - \Phi_{\sum |\{s|(s,r,o) \in \mathcal{L}'\}| = 0} \right) \quad (2)$$

Here, Φ_β is an indicator function returning 1 if the condition β holds, and 0 otherwise. $\mathcal{L}'$ denotes the event chain set and λ is a parameter.

For isolated predicates without historical support, we simulate dependencies using a binary indicator and incorporate external policy triples extracted from the "Emergency Response Plan Management Regulations" issued by the State Council of China. We

denote the policy triple set as $\mathcal{M} = \{m_1, m_2, \dots, m_n\}$, where each $m_i = (s_i, r_i, o_i)$ represents a resolution strategy.

Given a query $q = (s, r, ?)$, we define its historical event set $D^{(s,r)}$ and historical entity set $H^{(s,r)}$ as follows:

$$D^{(s,r)} = \bigcup \{(s, r, o)\} \quad (3)$$

$$H^{(s,r)} = \{o \mid (s, r, o) \in D^{(s,r)}\} \quad (4)$$

Entities not in $H^{(s,r)}$ are considered non-historical. Similarly, for each triple in the policy set, we evaluate whether it relates to the subject and predicate of the query, and define:

$$M^{(s,r)}(o) = \mu \left(\Phi_{\sum |\{s|(s,r,o) \in \mathcal{M}\}| > 0} - \Phi_{\sum |\{s|(s,r,o) \in \mathcal{M}\}| = 0} \right) \quad (5)$$

where μ is a parameter. The resulting frequency-based vectors integrate historical and non-historical information (policy information) to represent query-specific triple-level dependencies.

2.4 Event Chain Evolution Pattern

The event chain evolution module aims to enhance query chain representation by leveraging structurally and semantically similar event chains.

2.4.1 Similar Event Chain Matching. To fully explore information that shares similar evolution patterns with the query event chain in the dataset, we construct a set of similar event chains through the following steps:

- **Key Predicate Matching:** Define an indicator a , where $a = 1$ if there exists a triple matching the query pair (s, r) in another event chain, and $a = 0$ otherwise.
- **Semantic Similarity:** Transform the query event chain into a textual sequence, encode it using BERT, and compute the cosine similarity of semantic vectors as the similarity score, denoted as parameter b .
- **Structural Similarity:** Model the structure of the query event chain using GCN and apply the MVGRL model [8] for contrastive learning of entities (nodes) and relations (edges), resulting in a structural similarity score denoted by parameter c .

By combining these three aspects, the similarity between event chain i and chain j , denoted as sim_{ij} , is defined as:

$$\text{sim}_{ij} = \lambda_1 a_{ij} + \lambda_2 b_{ij} + \lambda_3 c_{ij} \quad (6)$$

where λ_1 , λ_2 , and λ_3 are weight hyperparameters.

Taking the predicate of the triple with the missing entity as the core signal, we integrate the chain structure and use BERT to encode the entire chain. Based on the combined similarity from key predicate, structure, and semantics, we select the top k most similar event chains to the query chain C_i , forming the similar event chain set:

$$S_i = (C_{i1}^{\text{sim}}, C_{i2}^{\text{sim}}, \dots, C_{ik}^{\text{sim}}) \quad (7)$$

2.4.2 Event Component Relation Matrix. Based on the similar event chain set S_i , we extract event evolution patterns and construct event evolution matrices to model transition information, as well as role co-occurrence relations between entities (as subject/object) and predicates. Specifically:

- **Predicate Transition Matrix:** Based on the predicate set $\mathcal{R}$, we construct a predicate transition matrix $\mathbf{M}_r \in \mathbb{R}^{|\mathcal{R}| \times |\mathcal{R}|}$, where $\mathbf{M}_r(u, v)$ represents the frequency of predicate u being followed by predicate v within event chains.
- **Subject-Predicate Association Matrix:** Based on entity set $\mathcal{E}$ and predicate set $\mathcal{R}$, we construct a subject-predicate matrix $\mathbf{M}_{sr} \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{R}|}$, where $\mathbf{M}_{sr}(e, r)$ denotes the frequency of entity e appearing as the subject with predicate r .
- **Object-Predicate Association Matrix:** Similarly, we construct an object-predicate matrix $\mathbf{M}_{or} \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{R}|}$, where $\mathbf{M}_{or}(e, r)$ indicates the frequency of entity e serving as the object with predicate r .

By integrating the above three matrices $\mathbf{M}_r$, $\mathbf{M}_{sr}$, and $\mathbf{M}_{or}$, we model the evolution information of the query event chain C_i , guiding downstream reasoning and completion tasks.

2.5 Hierarchical Association Rule Extraction

The hierarchical association rule extraction module aims to systematically construct a hierarchical rule system by incorporating event dependency patterns and event chain evolution knowledge. These rules are used to assist large language models in reasoning and entity completion tasks. The overall rule construction is conducted at two levels: entity-level and concept-level.

2.5.1 Entity-Level Rule Set. We employ event dependency patterns and event chain evolution knowledge to extract entity-level rules. The rule extraction strategies are defined as:

1) For historical dependency rules, we extract high-frequency co-occurrence patterns between subject/object entities and various predicates from historical events to construct triple-based historical dependency rules. When the frequency-based score satisfies $Z^{(s,r)}(o) > \beta$, the corresponding rule is retained in the entity-level rule set, where β is a rule confidence threshold.

2) For non-historical dependency rules, we search the triple in the policy set to match the given subject and predicate. Thus, the triple is directly included in the entity-level rule set if $M^{(s,r)}(o) > \beta$.

3) For evolution-based rules, we first identify predicate pairs with high co-occurrence frequency using the predicate transition matrix $\mathbf{M}_r$. We then trace entity pairs involved in these transitions and form evolution-path-based rules. For a query q with known subject s and predicate r , we:

- Use $\mathbf{M}_{or}$ to find the most frequent object entities for predicate r ;
- Use $\mathbf{M}_{sr}$ and $\mathbf{M}_{or}$ to find which predicates these object entities frequently appear with as subjects or objects;
- Use $\mathbf{M}_r$ to identify predicates that frequently co-occur with the current predicate;
- Search within the current event chain for matching or semantically similar predicates.

Entities that meet the structural co-occurrence patterns and appear frequently in the identified predicate contexts are assigned higher confidence and extracted as candidate target entities in triple format. Ultimately, the entity-level rules are stored as triples (s, p, o) , where p denotes a predicate pattern, along with a confidence score $C_{\text{triple}}(s, p, o)$, to support ranking during inference.

2.5.2 Concept-Level Rule Set. To enhance the generalization capability of the rule system, we extend rule construction from the entity level to the concept level. First, we map entities in the event chains to standardized concept categories using external knowledge bases such as Wikidata. The mapping prioritizes canonical concept classes when available. For entities that fail to match or contain ambiguity, we utilize the DeepSeek-V3 language model API for semantic inference, followed by manual verification and correction of low-confidence inferences to obtain a precise and comprehensive entity-to-concept mapping table.

Based on this mapping, we perform concept substitution on the original event chains to replace entity mentions with their corresponding concepts. This abstraction allows for the derivation of more general logical rules that are not tied to specific entities. We then reapply the rule extraction strategies defined at the entity level—such as high-frequency co-occurrence patterns between subject/predicate/object triples and frequent predicate transition paths—to the concept-replaced chains. In particular, for each concept triple (c_s, r, c_o) , where c_s and c_o are the mapped concepts of subject and object, respectively, we calculate its confidence score C_{concept} as:

$$C_{\text{concept}}(c_s, r, c_o) = \frac{f(c_s, r, c_o)}{\sum_{c'_o} f(c_s, r, c'_o)} \quad (8)$$

where the numerator is the frequency of the triple, and the denominator is the total frequency of all object concepts c'_o given the same subject concept c_s and predicate r . Only rules satisfying $C_{\text{concept}}(c_s, r, c_o) > \hat{\beta}$ are retained in the concept-level rule set.

Table 1: Statistics of Different Types of Rules

Rule Type	SocialChains	ICEWSChains14	ICEWSChains05-15
Historical Dependency Rules (A)	1542	3938	10821
Strategic Rules (B)	1387	0	0
Evolution Rules (C)	24516	3818	12921
Concept-level Rules (D)	7966	166	204
Entity-level Rules (A+B+C)	27445	7756	23742
Total Rules (A+B+C+D)	35411	7922	23946

The resulting concept-level rules are stored as triples (c_s, p, c_o) along with their confidence scores $C_{\text{concept}}(c_s, p, c_o)$. These rules complement entity-level rules by offering abstracted logical guidance, thereby improving robustness and adaptability in reasoning tasks under unseen or sparse entity conditions.

2.5.3 Comprehensive Rule Set. The comprehensive rule set is constructed by integrating entity-level and concept-level rules. As shown in Table 1, the rules are derived from different sources and vary in volume across datasets. All rules share a unified storage structure consisting of rule type (entity-level or concept-level), head entity, predicate, tail entity (or concept), and confidence score. These rules are persisted in a lightweight JSON format for efficient dynamic loading and retrieval during both training and inference.

This rule construction process systematically captures local dependencies and global evolution patterns in event chains, significantly enhancing the model’s capacity for modeling and reasoning over complex event developments in the entity prediction task.

2.6 Training and Inference

To fully exploit rule-based evolution knowledge and the generative capabilities of large language models, we formulate the event chain completion task as an instruction-based question answering problem. We adopt the LoRA (Low-Rank Adaptation) technique to adapt ChatGLM3-6B through lightweight fine-tuning.

2.6.1 Input Construction and Data Organization. Each training and inference sample consists of two parts: 1) **Prompt:** The input prompt comprises the preceding $n-1$ observed events and the query triple $(s, r, ?)$, augmented with relevant rules. These rules are incorporated through soft prompting and rule annotation. Specifically, matched rules are embedded into the prompt as natural language hints, while key elements such as entities, predicates, and rule alignments are tagged within the sequence to enhance structural awareness. This instruction-style prompt guides the model to understand event progression, recognize the prediction target, and adhere to rule constraints. 2) **Target:** The correct answer entity is provided in a structured natural language form, serving as the reference output for training. After rule augmentation, all data samples are organized as (Prompt, Target) pairs and split into training, validation, and test sets to ensure task consistency and fair generalization evaluation.

2.6.2 LoRA Fine-tuning Strategy. Considering the high computational cost and overfitting risks of full fine-tuning LLMs, we adopt LoRA for efficient adaptation. We freeze most of the original model’s parameters and introduce low-rank trainable matrices to select

weight layers. During training, only the LoRA-specific parameters are updated, enabling the model to rapidly adapt to the event chain completion task while preserving prior knowledge. LoRA achieves task-specific adaptation with minimal training overhead, striking a balance between inference quality and fine-tuning efficiency.

2.6.3 Loss Function. We employ a standard language modeling loss. Specifically, the model generates an output sequence conditioned on the prompt, and the loss is calculated as token-level cross-entropy with the reference sequence:

$$\mathcal{L} = - \sum_{t=1}^{\hat{T}} \log P(y_t | y_{<t}, \text{Prompt}) \quad (9)$$

where $\hat{T}$ denotes the length of the target sequence, and $P(y_t | y_{<t}, \text{Prompt})$ is the probability of generating token y_t given the previous tokens and input prompt. This loss formulation allows the model to learn entity completion under a natural language generation paradigm, moving beyond traditional classification-based objectives.

2.6.4 Inference and Prediction. During inference, the model generates a candidate set of entities $O_c = \{o_1^c, o_2^c, \dots, o_k^c\}$ based on the input query and rule-enhanced prompt. The final prediction selects the top- k entities as:

$$\hat{o}_n \in \text{TopK}(\text{LLM}(q, \mathcal{R}_{\text{rule}})) \quad (10)$$

The predicted entities are expected to satisfy both historical event progression and rule-based constraints, thereby improving both the accuracy and interpretability of the entity completion task.

3 Experiments

3.1 Dataset

As there is currently no publicly available benchmark for the event chain entity completion task, we construct three datasets, such as SocialChains, ICEWSChains14, and ICEWSChains05-15 datasets, to evaluate our proposed model and facilitate comparisons with existing baselines. These datasets are derived from the Chinese THUCNews corpus and the English ICEWS knowledge graph datasets.

SocialChains: The THUCNews corpus, compiled by Tsinghua University, contains over 740K news articles published between 2005 and 2011 across 14 categories. To construct structured event chains, we select documents from the "Society" category, which frequently exhibit coherent subject-action-object patterns (e.g., "suspect-attacked-victim" followed by "police-arrest-[X]"). These

Table 2: Dataset Statistics

Dataset	#Chains	#Triples	#Entities	#Predicates	Train	Val	Test
SocialChains	30,389	349,670	282,214	53,897	18,233	6,078	6,078
ICEWSChains14	9,183	29,880	1,124	223	5,509	1,837	1,837
ICEWSChains05-15	32,336	83,925	2,625	235	19,402	6,467	6,467

narratives are particularly suitable for modeling causality and temporal continuity. After preprocessing steps including sentence segmentation, event extraction, coreference resolution, and entity normalization, we obtain a high-quality Chinese event chain dataset, named SocialChains, characterized by rich entity and relation types, semantic coherence, and temporal structure.

ICEWSChains14 and **ICEWSChains05-15**: The ICEWS data is sourced from the Integrated Crisis Early Warning System and contains structured geopolitical events. We use the ICEWS14 and ICEWS05-15 subsets, which span 2014 and 2005–2015 respectively. Each event is represented as a quadruple (s, r, o, t) , with timestamp granularity of one day. We process these into English event chains named ICEWSChains14 and ICEWSChains05-15 by linking temporally and semantically adjacent events. A three-level quality control strategy focusing on entity consistency, chain continuity, and semantic cohesion is applied. We split each dataset into training/validation/test sets in a 6:2:2 ratio. Detailed statistics are summarized in Table 2.

3.2 Baseline Methods

We compare ReRule against a range of representative models, including static knowledge graph completion methods, such as DistMult [36], RGCN [26], TransE [1] and SACN [27], temporal knowledge graph completion methods, such as RE-GCN [16], CyGNet [38], CNE [15], CENET [34] and xERTE [7], and large language models, such as ChatGLM3 [6] and T5 [23].

For each baseline model, we adopt the hyperparameter settings reported in their original papers whenever applicable. However, due to the lack of prior work that exactly matches the setting of our event chain completion task, we adapt the input and output formats of all baseline models to support event chain data and produce predictions consistent with our task formulation. Additionally, we ensure that all models are trained and evaluated under comparable computational constraints, keeping core architectural components and key training parameters aligned for fair comparison. We compare ReRule with all baseline models using four standard metrics: MRR, Hits@1, Hits@3, and Hits@10.

3.3 Performance Comparison

To fairly and accurately evaluate the performance improvements of the proposed ReRule model, we conduct experiments following a standardized procedure on three constructed datasets: SocialChains, ICEWSChains14, and ICEWSChains05-15. The results are summarized in Table 3. Bold highlights the best performance, while underline denotes the second best.

Static knowledge graph completion models (e.g., DistMult, RGCN, TransE, SACN) perform poorly across all three datasets, with MRR scores consistently below 0.25. This suggests that static embedding-based models, which lack temporal and sequential modeling, are

inadequate for capturing the causal and temporal dependencies inherent in event chains. Their limitations are particularly evident on the SocialChains dataset, where performance approaches unusable levels, indicating that simple static embeddings are insufficient for complex event chain prediction tasks.

Temporal knowledge graph completion models (e.g., RE-GCN, CyGNet, CNE, CENET, xERTE) demonstrate moderate improvements by leveraging temporal and structural information in graphs. Compared to static baselines, these methods show clear gains on Hits@10 in the SocialChains dataset. However, their overall MRR remains low, suggesting that these models struggle to capture fine-grained chain-level dependencies. On the ICEWSChains14 and ICEWSChains05-15 datasets, RE-GCN, CENET, and xERTE achieve competitive results. It is because the extracted chain structures in these datasets preserve much of the original ICEWS event distribution. Furthermore, these methods benefit from architectural similarities between event chain modeling and conventional temporal KGC. Nonetheless, their lack of explicit chain-level reasoning still limits performance on sparse or structurally complex datasets, such as SocialChains.

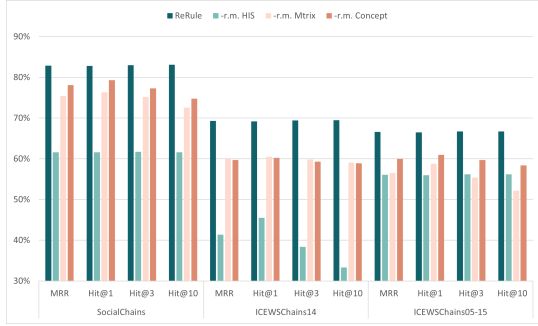
Despite not being specifically designed for event chain completion, the T5 model captures sequence-level patterns more effectively than a finetuned ChatGLM3. Notably, ChatGLM3 performs poorly under direct instruction-based finetuning, indicating that this task poses non-trivial challenges for general-purpose LLMs and demands task-specific adaptation strategies.

The ReRule model further incorporates a hybrid mechanism of rule retrieval and large language model generation, achieving significant performance. On the SocialChains dataset, ChatGLM3+ReRule outperforms LLaMA3+ReRule across all metrics, achieving higher MRR and Hit@10 than all other baselines. This demonstrates its superior generalization and reasoning capabilities for Chinese social event chains. On the ICEWSChains14 and ICEWSChains05-15 datasets, while LLaMA3+ReRule achieves higher scores on Hit@3 and Hit@10, ChatGLM3+ReRule performs better on MRR and Hit@1, showing its adaptability and robustness in large-scale temporal event reasoning tasks.

Due to their design characteristics, LLaMA3 generally performs better on English datasets, while ChatGLM3 is more suited for Chinese. Overall, ChatGLM3+ReRule shows higher stability and better comprehensive performance, making it more suitable as the backbone model for the ReRule framework. Although LLaMA3+ReRule demonstrates competitiveness in some metrics, ChatGLM3+ReRule proves to be a more balanced choice across diverse event chain scenarios.

Table 3: Comparison of ReRule and baseline models on SocialChains, ICEWSChains14, and ICEWSChains05-15 datasets.

Model	SocialChains				ICEWSChains14				ICEWSChains05-15			
	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10
DistMult	0.015	0.015	0.015	0.015	0.148	0.070	0.168	0.302	0.138	0.072	0.151	0.240
RGCN	–	–	–	–	0.001	–	–	0.001	0.003	–	0.004	0.004
TransE	0.018	0.010	0.022	0.035	0.182	0.021	0.278	0.481	0.209	0.040	0.316	0.511
SACN	0.015	0.015	0.015	0.015	0.185	0.090	0.215	0.386	0.234	0.130	0.274	0.434
RE-GCN	0.016	0.001	0.004	0.105	0.367	0.266	0.412	0.553	0.381	0.276	0.432	0.584
CyGNet	0.081	0.060	0.092	0.120	0.351	0.257	0.394	0.532	0.368	0.273	0.412	0.545
CNE	0.202	0.187	0.212	0.221	0.327	0.235	0.368	0.506	0.374	0.271	0.420	0.575
CENET	0.124	0.093	0.135	0.173	0.396	0.293	0.449	0.585	0.431	0.325	0.486	0.629
xERTE	0.193	0.179	0.201	0.217	0.408	0.327	0.457	0.573	0.466	0.378	0.523	0.639
ChatGLM3 (Fine-tuned)	0.105	0.105	0.105	0.105	0.100	0.096	0.104	0.108	0.060	0.055	0.062	0.070
T5 (Fine-tuned)	0.326	0.280	0.360	0.417	0.101	0.062	0.114	0.215	0.097	0.056	0.113	0.209
LLaMA3+ReRule	<u>0.795</u>	<u>0.805</u>	<u>0.822</u>	<u>0.829</u>	<u>0.641</u>	<u>0.512</u>	0.746	0.797	<u>0.594</u>	<u>0.479</u>	0.676	0.767
ChatGLM3+ReRule	0.829	0.828	0.830	0.831	0.693	0.692	<u>0.694</u>	<u>0.695</u>	0.666	0.665	<u>0.667</u>	<u>0.667</u>

**Figure 3: Ablation results of ReRule on SocialChains, ICEWSChains14, and ICEWSChains05-15 datasets.**

3.4 Ablation Study

To analyze the contribution of each component in ReRule, we constructed three ablated variants by removing or replacing key modules. Their results on three datasets are shown in Figure 3 in terms of MRR and Hits@K.

- **-r.m. HIS:** This variant removes the event-level dependency rules (HIS) module from ReRule. During retrieval, it ignores rules associated with entity-level historical interactions, including strategy rules from external policy sets, thereby discarding triple-level historical knowledge.
- **-r.m. Matrix:** This variant excludes the evolutionary rules (Matrix) module. It ignores the evolutionary rules embedded in the entity layer of the dual-layer rule base, thereby removing knowledge of event chain evolution.
- **-r.m. Concept:** This variant removes the concept-level rule (Concept) module. During retrieval, it excludes abstract matching rules from the concept layer, discarding conceptual abstraction knowledge.

From the overall trend, removing any rule module significantly reduces model performance, confirming the importance of all three rule types in the event chain prediction task. Specifically, removing event-level dependency rules leads to a substantial drop in all

metrics on the SocialChains dataset, indicating the critical role of past interactions in entity prediction. The removal of evolutionary rules results in the largest performance degradation across all datasets, highlighting the importance of modeling event transitions and temporal logic. Lastly, although the removal of the concept-level module leads to relatively smaller performance drops, it still plays an auxiliary role by mapping entities to abstract categories, reducing the prediction space, and improving semantic consistency.

4 Case Study

To intuitively illustrate the overall design and implementation of the ReRule model and to investigate the effectiveness of rule-guided mechanisms in enhancing large language model reasoning, we conduct a detailed case study on a representative event chain.

Figure 4 presents the construction of similar event chains during the rule-based event evolution process. The case centers around an incident involving a collapse in the Nanning Metro Line 1 at the Guomao Center, followed by related responses and interventions. The first row of the figure indicates the current query chain, while the others are retrieved based on a composite similarity metric that integrates key predicates, structural patterns, and semantic embeddings. Lighter cells indicate lower similarity scores, and stars denote the most similar chain in each comparison. From the visualization, we observe that the retrieved chains share key elements with the query chain, such as entities like construction companies or monitoring and early warning systems. Chains involving similar predicates like “strengthen”, especially in the context of events leading to property damage or injury, tend to be ranked higher. Objects in these similar chains are extracted as examples during the rule mining process, and once their frequency surpasses a predefined threshold, they are included in the corresponding rule set.

Figure 5 shows how rule matching, prompt generation, and prediction are performed. During prompt construction, the query triple is matched against various rule templates. First, at the concept level, the system retrieves rules associated with the predicate “strengthen”, identifying its object type as predominantly Action. Then, at the entity level, the triple (supervision unit, strengthen, ?)

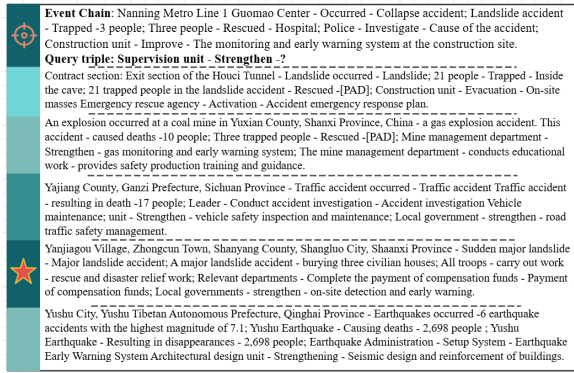


Figure 4: Similar event chain construction during the rule-based event evolution process

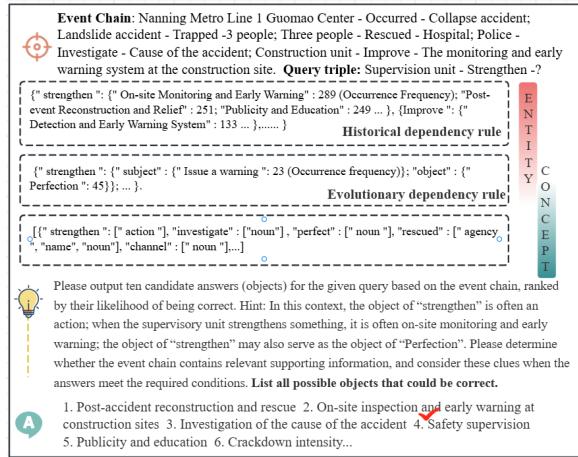


Figure 5: The rule-guided prompt generation and model prediction.

is matched against the rule set to retrieve statistically frequent objects. If the predicate occurs frequently enough, finer-grained rules involving subject-predicate pairs are also considered. Lastly, the query triple is evaluated in the context of previous event chains to identify whether its object co-occurs with other roles elsewhere in the chain (e.g., as a subject or object of another event). This hierarchical rule-matching process yields a structured prompt that serves as input for the LLM during fine-tuning. The generated answers are shown in the lower part of Figure 5. The top-ranked prediction is post-accident reconstruction and rescue, which—while frequently co-occurring with the predicate “strengthen”—typically appears with governmental or emergency response subjects, making it a suboptimal answer in this context. The second prediction, monitoring and early warning system, directly aligns with an earlier fact in the event chain—construction company—improve—monitoring system—and exhibits strong contextual and causal coherence. This indicates a correct inference. The third answer, accident cause investigation, is a partial match derived from related rules such as investigate—accident. However, the subject “supervision unit” is

misaligned, as such an action would more plausibly be undertaken by the police. Other predictions, like safety supervision, publicity and education, and crackdown intensity, reflect semantic relevance but fail to leverage the specific context of the input chain, thus being ranked lower.

This case study demonstrates the effectiveness of rule-guided prompting in enhancing LLM reasoning over structured event chains. By leveraging statistical constraints, event evolution rules, and concept-level semantic filters, ReRule facilitates a shift from mere linguistic association toward more knowledge-grounded inference, thereby improving semantic precision and reasoning interpretability in entity prediction tasks.

5 Conclusion

This paper proposes ReRule, a novel model for entity completion in event chains by integrating multi-granularity rule sets with pretrained language models. ReRule leverages structural insights from event chain data and designs a hierarchical rule construction mechanism at the entity, triple, and chain levels. These rules form a structured knowledge base that captures both conceptual abstraction and instance-level semantics. Built upon this, ReRule incorporates the rule set into a language model through instruction tuning, using soft prompts and explicit annotations to guide prediction. Additionally, a cooperative inference mechanism based on rule matching and language model scoring enhances prediction accuracy and robustness. Experimental results demonstrate that ReRule consistently achieves state-of-the-art performance across multiple datasets and metrics, exhibiting strong interpretability and generalization capability.

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A Related Work

This section provides a comprehensive review of representative works in temporal knowledge graph completion and event chain prediction.

A.1 Temporal Knowledge Graph Completion

TKGC extends static KGC by incorporating temporal information into knowledge modeling. Time-aware extensions of translational models, including TTransE [11] and HyTE [4], introduce time embeddings or project entities into temporal hyperplanes. ConT [19] and TNTComplEx [13] treat time as a continuous variable and employ time-specific tensor decomposition. GNN-based methods have been adapted for TKGC as well. ExKGR [35] designs event-aware attention over dynamic subgraphs. BoxTE [21] leverages box embeddings to represent temporal intervals and reason over temporal scopes. Neo-TKGC [22] leverages both graph structure and temporal dependencies to enhance entity and relation representation. TeRDy [17] employs time-invariant embeddings and temporally dynamic embeddings to capture temporal relational dynamics. Some approaches further introduce logical constraints or temporal patterns to enhance interpretability and generalization. For example, CyGNet [38] encodes periodic temporal dependencies, while ChronoR [25] models relative temporal relations among entities and events. Despite these advancements, existing TKGC methods still face challenges in capturing long-range dependencies,

reasoning across multiple event chains, and handling ambiguous temporal contexts.

A.2 Script Event Prediction

Script event prediction focuses on inferring the most plausible subsequent event based on a given sequence of prior events. Unlike structured knowledge graphs, this task often operates in unstructured narrative spaces, where events are described in natural language. It is closely related to temporal reasoning and complements TKGC by emphasizing causal and sequential patterns within event chains. Early methods adopt event-pair models, which rely on co-occurrence statistics to capture local dependencies. Chambers and Jurafsky [2] introduce narrative chains using PMI to learn prototypical event transitions. Subsequent works [10, 24] apply skip-gram-style models to estimate event-relatedness. However, such approaches are limited in modeling complex or long-range event structures. To address this, event-chain models were proposed to capture sequence-level patterns. Hu et al. [9] employ hierarchical LSTMs to encode event and word sequences jointly. Lv et al. [18] use attention to model inter-event dependencies. These models enhance temporal understanding but still struggle with non-linear structures. More recent event-graph models represent events as nodes connected via dependency edges, enabling richer structural reasoning. Li et al. [14] build narrative event graphs and applied GNNs for context-aware representation. Wang et al. [31] combine GCNs and sequential encoders for improved representation learning. SSD-GM [33] employs event-centric structure information based on GNN and sentiment information to enhance the generative model. External knowledge models introduce background information to compensate for data sparsity. Zhou et al. [37] utilized the TimeTravel dataset and designed self-supervised pretraining tasks for enhanced generalization. Wei et al. [32] employ the intrinsic knowledge of LLMs for script event prediction by using event evolution patterns. Furthermore, DocSEP [20] is designed to learn sequential ordering between events in long-form documents. In addition, recent advances in adversarial retrieval [30] could be used to archive and align global event knowledge, thus providing rich evidence for event representation and reasoning. In summary, script prediction offers valuable insights for modeling temporal and causal dynamics in event chains, though challenges remain in generalization, structural reasoning, and knowledge integration.

B Dataset Construction Details

This section introduces the construction details of the novel event chain datasets, such as the Chinese dataset SocialChains and the English datasets ICEWSChains14 and ICEWSChains05-15. Table 4 presents a comparison of the number of event chains, average chain length, event chain integrity, and entity overlap rate on the three datasets.

B.1 SocialChains

The event chain dataset SocialChains is extracted from the THUCNews news dataset. First, the DeepSeek knowledge extraction tool is used to extract structured event triplets from each event case in the event case library of THUCNews. Then, a semantically coherent event chain is formed by organizing events in relative chronological

order and subject organization. Further, we establish a multidimensional verification mechanism and standardized rules to enhance the quality of datasets.

Chain extraction We employ the DeepSeek-V3 API as the basic tool to extract event triplets, such as (subject, predicate, object), by using the designed prompt templates. The extracted event triplets are ranked according to the relative time sequence to form the social event chains. The number of event chains after extraction is 40003 for THUCNews. Meanwhile, we eliminate the event chains with sensitive event types or keywords, redundant event descriptions, and the absence of core event entities.

Multidimensional verification Despite the reliability of the DeepSeek-V3 model in event extraction, the initial event chain extraction results nevertheless contain certain discrepancies and errors in event representation. By examining the distribution of the existing event chain across multiple dimensions, such as entity, chain structure, and semantics, we employ standard NLP techniques to clean and balance the dataset, thereby enhancing its overall quality. The implementation procedure is detailed below:

- **Entity distribution verification:** Entity-level verification is to conduct word frequency and part-of-speech statistics on entities and predicates contained in the event chain dataset. On one hand, the numerous monetary numerical entities in the dataset not only provide limited informational value for model learning, but can also divert the model from its intended learning objective due to their high frequency. Therefore, the regular filtering mechanism is used to clean triples in the data set, such as "accident - loss of -1000 yuan" and "Wallet - have -200 dollars". On the other hand, some event triple items are null during extraction, owing to some events with only subject and predicate structure, or unclear entity relationships. Thus, we employ the "[PAD]" symbol to occupy the missed items in the triplet for unifying the representation.
- **Chain distribution verification:** The verification at the chain level focuses on the length and integrity of the event chain in the event chain dataset. On one hand, the extracted event chain length is obviously unbalanced in the dataset. Thus, event chains that are limited in number but markedly longer than the average are deleted after manual assessment. Meanwhile, singleton event chains (length=1) should be discarded, as they do not constitute a meaningful sequence. On the other hand, the event chain with a high proportion of "[PAD]" is considered to have low event integrity. Therefore, cleaning the event chain with a high proportion of "[PAD]" can ensure the relative integrity of the event chain dataset.
- **Semantic distribution verification:** The Semantic distribution verification focuses on the entity overlap of the event chain in the dataset. The same entity can be represented differently both within and across event chains. This inconsistency leads to a meaningless proliferation of entity mentions in the set, which ultimately degrades the model's accuracy in entity prediction. Thus, we use the overlapping rate of entities between chains to represent the degree of semantic association between different chains in the dataset. The average overlapping rate between all chain pairs is defined

Table 4: Comparison of SocialChains, ICEWSChains14, and ICEWSChains05-15 datasets before and after processing.

	SocialChains		ICEWSChains14		ICEWSChains05-15	
	Before	After	Before	After	Before	After
Number of event chains	40003	30389	2821	9183	5202	32336
Average chain length	11.74	15.02	11.25	16.54	11.24	16.21
Event chain integrity	99.20%	99.40%	100%	100%	100%	100%
Entity overlap rate	0.25%	0.28%	0.43%	1.89%	0.48%	0.87%

as:

$$EntityOverlapRate = \frac{2}{N(N-1)} \sum_{1 \leq i \leq j \leq N} \frac{|E_i \cap E_j|}{|E_i \cup E_j|} \quad (11)$$

where E_i denotes the entity set in the i -th chain and N is the number of chains. Using this metric, we can gauge the structural density of an event chain—ranging from sparse and isolated to densely interconnected—and evaluate its level of informational coherence.

- **Manual verification:** To ensure the reliability of the data set, 5% of the event chain samples were randomly selected from the final event chain data set, and two graduate students with the natural language processing background independently completed the annotation evaluation task. According to the original text content, each evaluator judges the rationality of the structure of each event in the event chain, the accuracy of the triple [subject, predicate, object], and whether the internal logical coherence of the event chain is good. In order to ensure the objectivity and consistency of the assessment, a unified assessment guide was provided for the two graduate students before the assessment, including event extraction criteria, entity unification principle, reference resolution rules, etc. The evaluation results show that more than 80% of the event chains are evaluated as "good" or "excellent", which verifies the effectiveness of the automatic extraction and processing process.

B.2 ICEWSChains14 and ICEWSChains05-15

For standard temporal knowledge graph datasets like ICEWS14 and ICEWS05–15, the original data already comprises event triplets (subject, predicate, object) along with their corresponding timestamps. The primary processing objective is therefore to construct coherent event chains, strengthen their capacity for contextual narrative representation, and systematically organize and evaluate the data. The entire procedure adheres to the "entity-chain-semantics" triple quality control strategy—previously introduced for Chinese datasets—with the goal of enhancing structural rationality and semantic relevance. The specific workflow is outlined below:

Chain extraction Event chain extraction is to identify events with potential semantic links and connect them based on their timestamps to form a continuous chain. The specific steps are as follows:

- **Initial screening of triplets:** Sort the quadruple (s, p, o, t) in the original data by timestamp and unify the time granularity to be precise 'Day'; Simultaneously remove predicate terms with ambiguous meanings or significant statistical noise (such as "Unknown", "NA", etc.)

- **Central event selection:** Select triplets with moderate frequency of occurrence of subjects or objects in the data as central events to avoid model learning being dominated by high-frequency entities (such as "United States").
- **Chain construction based on adjacency extension:** Starting from a central event, a chain is constructed by chronologically ordering events from adjacent entities within a fixed time window (± 30 days), and is subsequently pruned to a reasonable length.

Multidimensional verification To ensure that the constructed chain has a high degree of rationality in both structure and semantics, the quality control mechanisms are as follows:

- **Entity frequency balance verification:** Count the global occurrence frequency of entities in the chain, filter out chains dominated by high-frequency entities, and alleviate only the high-frequency entity learning problem;
- **Chain distribution verification:** Exclude singleton event chains (length=1), sample chains with a reasonable length range (such as 2-25) according to the actual distribution, and control the long tail effect;
- **Semantic consistency verification:** As the original ICEWS series dataset has clear entity types, the relevant operations for semantic consistency verification are omitted.

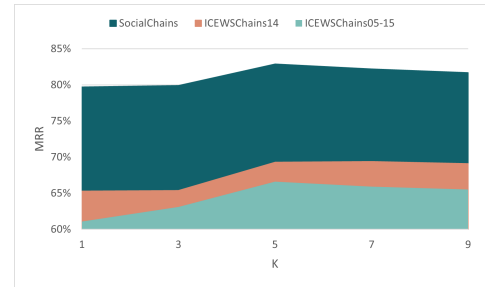


Figure 6: Impact of different k values (top- k similar event chains) on ReRule across three datasets

C Event Chain Number Effect

The number of retrieved similar event chains (k) plays a crucial role in enriching the model's input with historical event evolution patterns. To investigate how k influences performance, we fix the model structure and training procedure, and evaluate ReRule under different values of $k \in \{1, 3, 5, 7, 9\}$. The results on all three datasets are shown in Figure 6. We observe a consistent "rise-then-fall" pattern across all datasets: model performance improves as k increases

from 1 to 5, then declines when k exceeds 5. This suggests that $k = 5$ strikes the optimal balance between providing rich historical context and avoiding noise. When k is small (e.g., $k = 1$ or 3), the retrieved event chains are typically highly relevant but insufficient in number to capture diverse evolution trajectories. This leads to limited context and underutilization of useful patterns, resulting in lower MRR scores. In contrast, when k is large (e.g., $k = 7$ or 9),

although more chains are incorporated, the likelihood of semantic drift or structural inconsistency increases. This introduces noisy or misleading information, which distracts the model and reduces prediction accuracy. These findings highlight the importance of carefully selecting k to balance “information enrichment” and “semantic interference.” While an insufficient number of chains results in sparse signals, an excessive number dilutes the model’s focus.